

Online BookStore Analysis SQL Project Report

Prepared By

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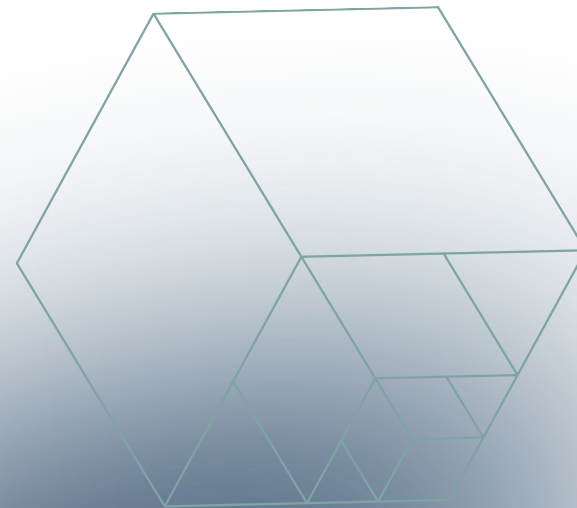


Project Overview

This project analyzes online bookstore sales data using SQL. The objective is to identify sales trends, customer behavior, top-performing books, and revenue insights through business analysis queries.

Tools Used

- PostgreSQL
- SQL
- Canva



Q1.What is the total revenue generated by the company?

Query

```
SELECT
  SUM(PAYMENT_AMOUNT) AS
TOTAL_REVENUE
FROM
  ORDERS;
```

Output

Data Output		Messages	Notifications
≡+ 📄 ▼ 📋 ▼ 🗑 🗄 ⬇ 📈 SQL			
	total_revenue numeric		
1	109456251.20		

Q2.What are the monthly sales and revenue trends?

Query

```
SELECT
  TO_CHAR(ORDER_DATE, 'Mon YYYY') AS MONTH,
  ROUND(SUM(PAYMENT_AMOUNT), 2) AS
MONTHLY_REVENUE
FROM
  ORDERS
GROUP BY
  MONTH
ORDER BY
  MONTH;
```

Output

17		Data Output		Messages	Notifications
≡+ 📄 ▼ 📋 ▼ 🗑 🗄 ⬇ 📈 SQL					
	month text		revenue numeric		
1	Apr 2024		2942397.2		
2	Apr 2025		2765458.8		
3	Apr 2026		3046599.5		
4	Aug 2023		3099952.5		
5	Aug 2024		2863040.0		
6	Aug 2025		2874532.3		
7	Dec 2023		3070874.7		
8	Dec 2024		2861224.6		
9	Dec 2025		3343131.3		
10	Feb 2024		2884851.3		
11	Feb 2025		2772608.0		
Total rows: 37		Query complete 00:00:00.137			

Q3.Which are the top 10 highest-selling products?

Query

```
SELECT
P.BOOK_TITLE,
SUM(O.QUANTITY) AS TOTAL_SOLD
FROM
ORDERS O
JOIN PRODUCTS P ON O.PRODUCT_ID = P.PRODUCT_ID
GROUP BY
P.BOOK_TITLE
ORDER BY
TOTAL_SOLD DESC
LIMIT
10;
```

Output

Data Output			Messages	Notifications
	book_title character varying (200)	total_sold bigint		
1	James Thomas	19		
2	Write.	17		
3	Pretty watch.	16		
4	Megan Brown	16		
5	Everly Walker	16		
6	Ok.	14		
7	Charles Moore	13		
8	Victoria Bennett	10		
9	Evening green computer.	10		
10	Athena Vasquez	10		
Total rows: 10			Query complete 00:00:00.195	

Q4.Which customers generated the highest revenue?

Query

```
SELECT
C.CUSTOMER_NAME,
ROUND(SUM(O.PAYMENT_AMOUNT), 2) AS TOTAL_REVENUE
FROM
CUSTOMERS C
JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID
WHERE
C.CUSTOMER_NAME != 'NA'
GROUP BY
C.CUSTOMER_NAME
ORDER BY
TOTAL_REVENUE DESC
LIMIT
10;
```

Output

Data Output			Messages	Notifications
	customer_name character varying (100)	total_revenue numeric		
1	Sheila Martinez	36800.07		
2	Amanda Harris	34976.64		
3	Daniel Hughes	28288.94		
4	Daniel Jones	27383.68		
5	Jason Lopez	27326.00		
6	Brian Brown	26962.87		
7	Joseph Walker	26224.43		
8	Michael Garcia	26112.41		
Total rows: 4695			Query complete 00:00:00.212	

Q5.Which product categories/genres perform the best?

Query

```
SELECT
P.GENRE,
ROUND(SUM(O.PAYMENT_AMOUNT), 2) AS TOTAL_REVENUE
FROM
ORDERS O
JOIN PRODUCTS P ON O.PRODUCT_ID = P.PRODUCT_ID
GROUP BY
P.GENRE
ORDER BY
TOTAL_REVENUE DESC;
```

Output

Data Output			Messages	Notifications
≡+  ▼  ▼     SQL				
	genre character varying (150) 🔒	highest_revenue numeric 🔒		
1	Sci-Fi	7651153.79		
2	Fiction	3745358.89		
3	Romance	3710242.22		

Q6.What is the average order value?

Query

```
SELECT
ROUND(AVG(PAYMENT_AMOUNT), 2) AS AVG_ORDER_VALUE
FROM
ORDERS;
```

Output

Data Output			Messages	Notifications
≡+  ▼  ▼     SQL				
	avg_order_value numeric 🔒			
1	7297.08			

Q7.How many orders are completed, pending, or cancelled?

Query

```
SELECT
  STATUS,
  COUNT(*)
FROM
  ORDERS AS TOTAL_ORDERS
WHERE
  STATUS IN ('Pending', 'Completed', 'Cancelled')
GROUP BY
  STATUS;
```

Output

Data Output		Messages	Notifications
			
			
	SQL		
	status character varying (100) 	count bigint 	
1	Completed	5905	
2	Cancelled	2968	
3	Pending	3035	

Q8.What is the cancellation rate of orders?

Query

```
SELECT
  CONCAT(COUNT(*) * 100 / 15000, '%') AS TOTAL_CANCELLATION
FROM
  ORDERS
WHERE
  STATUS = 'Cancelled';
```

Output

			
			
	SQL		
	total_cancellation text 		
1	19%		

Q9.Which products have the lowest stock and need restocking?

Query

```
SELECT
  BOOK_TITLE,
  STOCK
FROM
  PRODUCTS
WHERE
  STOCK > 0
  AND STOCK <= 10
ORDER BY
  STOCK ASC;
```

Output

	book_title character varying (200)	stock integer
1	Soldier know.	1
2	Across it music.	1
3	Success wind.	1
4	Drive control important.	1
5	Wind glass number.	2
6	Sense reason Democrat.	2
7	Fall reveal.	3
8	Style indicate house.	3
9	Industry seem beat.	4
10	Least weight south.	4
11	Send deep region.	4

Q10.Which customers are repeat customers?

Query

```
SELECT
  CUSTOMER_ID,
  COUNT(*) AS REPEATED_ORDER
FROM
  ORDERS
GROUP BY
  CUSTOMER_ID
HAVING
  COUNT(*) = 1;
```

Output

	customer_id integer	repeated_order bigint
1	11233	1
2	4790	1
3	3936	1
4	12502	1
5	5468	1
6	13093	1
7	13520	1
8	4326	1
9	2520	1
10	9038	1

Q11.Which month had the highest number of orders?

Query

```
SELECT
TO_CHAR(order_date, 'Mon YYYY') AS month,
COUNT(*) AS orders
FROM orders
GROUP BY month
ORDER BY orders DESC
LIMIT 1;
```

Output

	month text	orders bigint
1	Jul 2025	447

Q12.Which products are rarely ordered?

Query

```
SELECT
p.book_title,
SUM(o.quantity) AS total_sold
FROM orders o
JOIN products p
ON o.product_id = p.product_id
GROUP BY p.book_title
ORDER BY total_sold ASC
LIMIT 10;
```

Output

	book_title character varying (200)	total_sold bigint
1	Dominic Guzman	0
2	Interesting.	0
3	To.	0
4	Christopher Flores	0
5	Executive mission.	0
6	Ezekiel Jordan	0
7	Christian Fong	0
8	Care read would.	0
9	Angel Xiong	0
10	Ian Miller	0

Q13.What is the total quantity of products sold?

Query

```
SELECT
SUM(QUANTITY) AS TOTAL_PRODUCT_SOLD
FROM
ORDERS;
```

Output

Data Output		Messages	Graph Visualiser	×	Notif
≡+	📄	▼	📋	▼	🗑️
	total_product_sold				
	bigint				
1	68701				

Q14.What is the revenue contribution by each genre/category?

Query

```
SELECT
P.GENRE,
SUM(PAYMENT_AMOUNT) AS TOTAL_REVENUE
FROM
ORDERS O
JOIN PRODUCTS P ON O.PRODUCT_ID = P.PRODUCT_ID
WHERE
GENRE != 'NA'
GROUP BY
P.GENRE
ORDER BY
TOTAL_REVENUE DESC;
```

Output

160		Data Output		Messages	Graph Visualiser	×	No
≡+	📄	▼	📋	▼	🗑️	📊	SQL
	genre	total_revenue					
	character varying (150)	numeric					
1	Sci-Fi	7651153.79					
2	Fiction	3745358.89					
3	Romance	3710242.22					
4	History	3644392.52					
5	Fiction	3378551.97					

Q15.What are the top 5 most profitable products?

Query

```
SELECT
P.BOOK_TITLE,
SUM(O.PAYMENT_AMOUNT) AS TOTAL_REVENUE
FROM
ORDERS O
JOIN PRODUCTS P ON O.PRODUCT_ID = P.PRODUCT_ID
GROUP BY
P.BOOK_TITLE
ORDER BY
TOTAL_REVENUE DESC
LIMIT
5;
```

Output

174

Data Output Messages Graph Visualiser X Notifications

SQL

	book_title character varying (200)	total_revenue numeric
1	Jennifer Stevens	25690.70
2	Information couple.	25122.50
3	John Delgado	24436.71
4	Ok.	23725.41
5	Charles Moore	23131.04

Key Business Insights

- Certain book genres generated higher revenue.
- Monthly sales showed fluctuations across different periods.
- Some books were rarely ordered and may require promotion.
- A few customers contributed significantly to total revenue.
- Low-stock products may need restocking.
- Order cancellations can impact overall business performance.



Conclusion

- This SQL project helped analyze online bookstore sales data and generate meaningful business insights using real-world queries.
- The analysis improved understanding of customer behavior, revenue trends, and product performance.



Thank You!

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